EveSense

Catching an eavesdropper on a BB84 link that is already noisy

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The question

BB84 security rests on one fact: measuring a qubit in the wrong basis disturbs it. An intercept-resend attacker inserts errors, the two ends read those errors as QBER, and above roughly 11% QBER the link stops producing usable key.

Deployed fiber is never clean. Raman scattering, dark counts and polarization drift put real links at a few percent QBER with nobody listening. So the receiver is not comparing a measurement against zero. It is comparing against a baseline that moves.

Eve knows the baseline. A patient attacker keeps her interception rate under the noise floor. The receiver has one number and one threshold, and has to decide whether a rise in QBER is Eve or the weather.

What we built

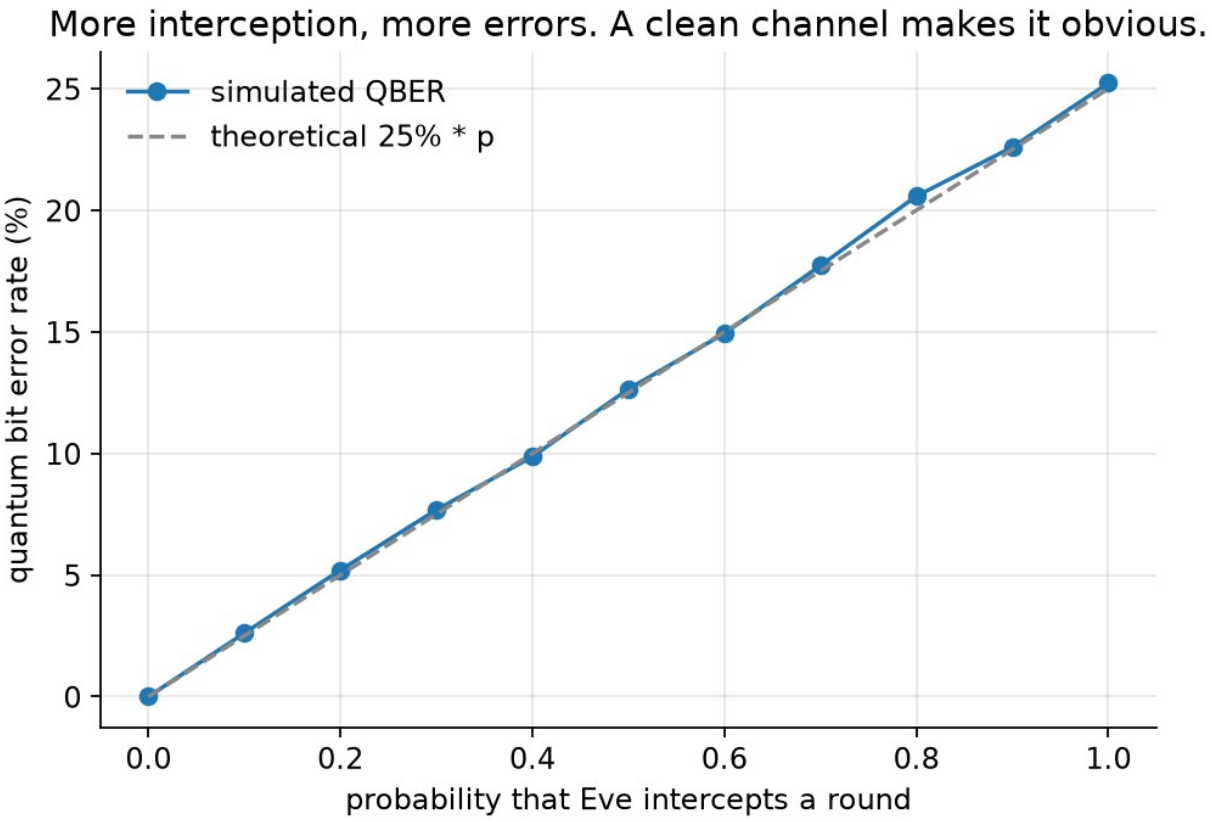
A BB84 link in Qiskit with a tunable attacker and a tunable channel, then a detection study across nine conditions.

One round is one qubit: Alice prepares |0>, |1>, |+> or |->, Eve intercept-measures and resends when she is active, the channel adds depolarizing and readout noise, Bob measures in the basis he picked.

Rounds are packed into batches of 64 qubits in a single stabilizer circuit, so 30000 rounds cost milliseconds. Eve's outcome is drawn from the Born rule, which is what a mid-circuit measure-and-resend gives.

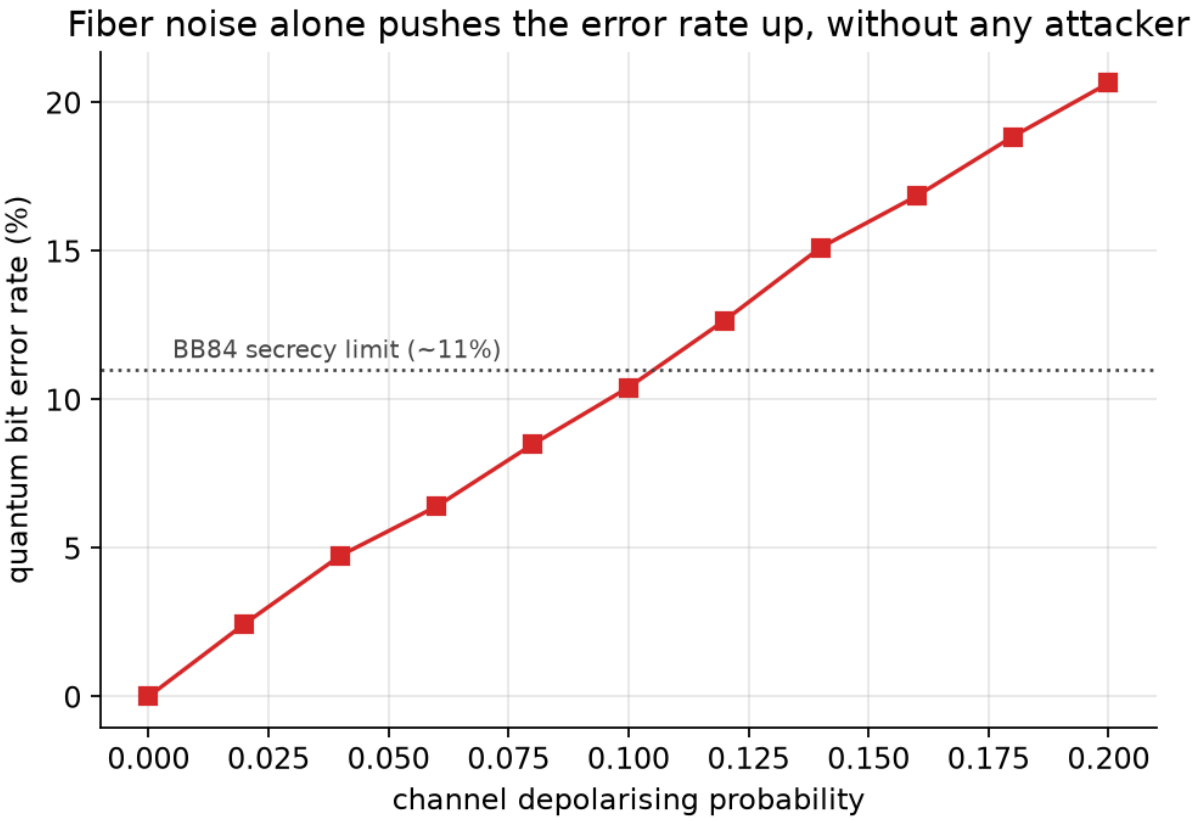
Four experiments: interception sweep, channel noise sweep, detection study with 12 repeats per condition, and the secure key rate curve.

Clean channel, tunable attacker



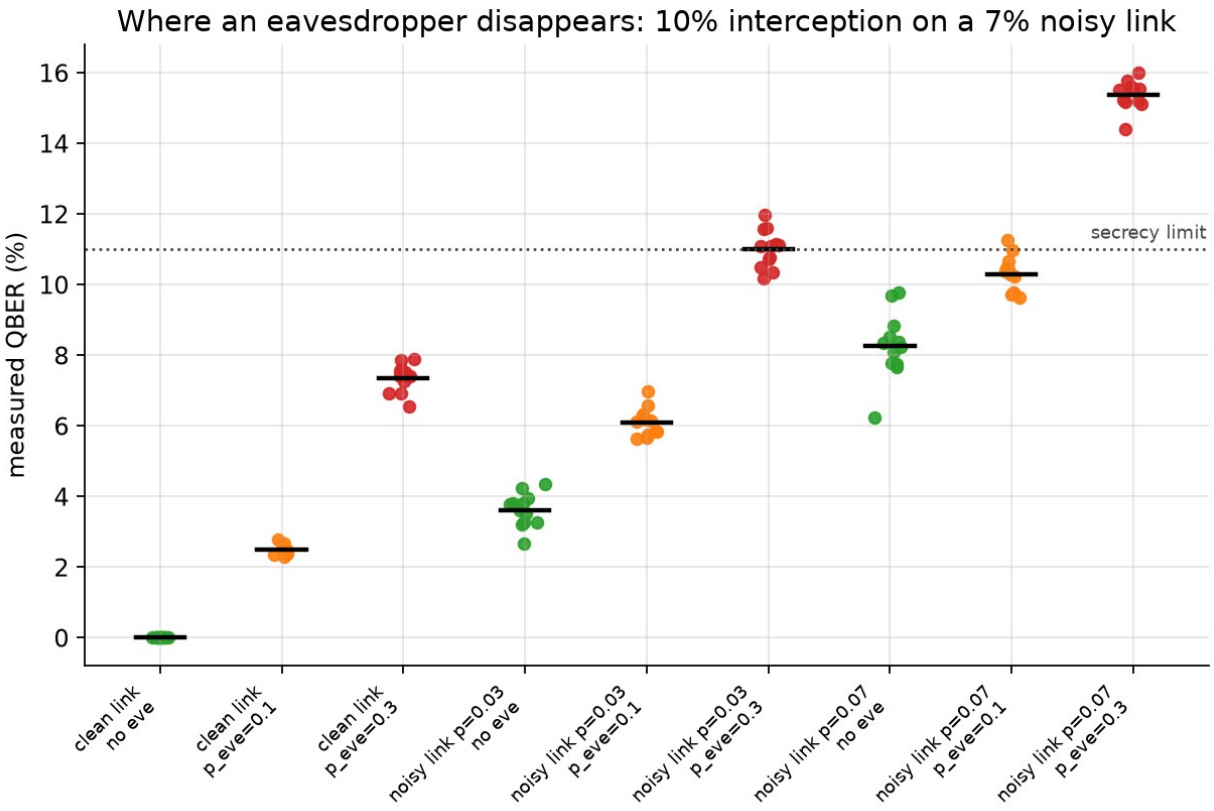
Interception sweep. The simulated points fall on the analytical 25% * p line, which is the sanity check every QKD demo should pass before claiming anything.

The baseline she hides in



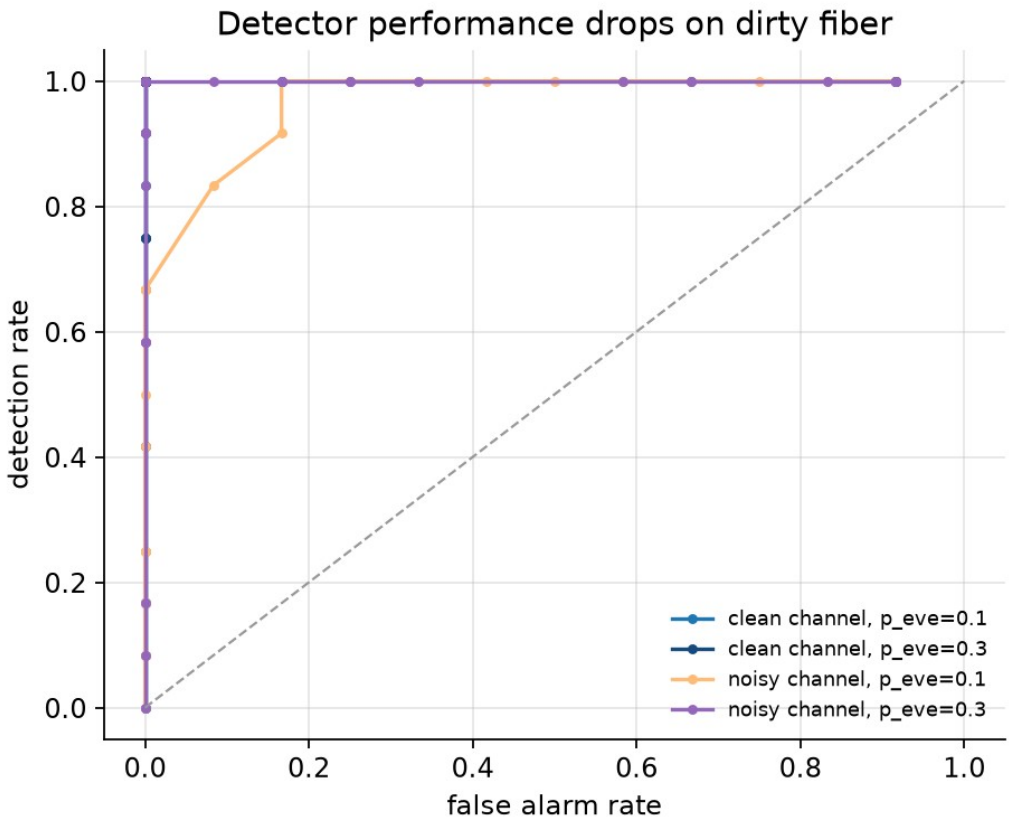
Channel noise with no attacker. By 12% depolarising probability the link crosses the 11% secrecy limit on its own.

Nine conditions, twelve runs each



QBER distributions per condition. The overlap between a tapped noisy link and an untapped noisy link is the whole problem.

Detection cost



Hit rate against false alarm rate. The noisy-channel curves fall toward the diagonal exactly where the attacker is quietest.

What the numbers say

On a clean link the protocol behaves exactly as theory predicts: QBER tracks 25% of the interception probability, and the measured curve sits on the analytical one.

A 7% depolarizing channel on its own produces about 7% QBER. That is the same order of magnitude as a 10% interception rate on clean fiber. The receiver cannot tell those two apart from a single measurement.

The ROC curves show the tradeoff. On clean fiber almost any threshold separates tapped from untapped. On dirty fiber the curve collapses toward the diagonal at low interception rates, which is where a patient attacker works.

Past about 11% QBER error correction and privacy amplification take everything. The link stays up and stops being worth anything.

Limits and next steps

Eve here is a textbook intercept-resend attacker. Photon-number- splitting and coherent attacks are not in the model, and nothing in this work is a security proof.

The channel is depolarizing plus symmetric readout noise. Real fiber drifts in polarization and loses differently per wavelength.

The detector is a single threshold. A real receiver discloses part of the key, estimates QBER on that subset and adapts the threshold against a measured baseline instead of a fixed number.

The next version of this work would estimate the baseline online and run the detection on the residual, which is the version that would survive contact with a real link.

Disclosure

This project was developed with the assistance of generative AI tools for research, structuring and content generation. The final content has been reviewed and validated by the author.

Full compliance with: Reg. (EU) 2024/1689 (AI Act) Art. 50 Italian Law 132/2025 Art. 13 EU Code of Practice on AI Transparency (2026) U.S. AI Labeling Act (S. 4915) - U.S. H.R.8479 U.S. Executive Order on AI (Dec 2025) California TFAIA & AB 2013 (eff. Jan 2026)